

NOTE

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NOTE

Inter-site generalizability of EEG based age prediction algorithms in the preterm infant

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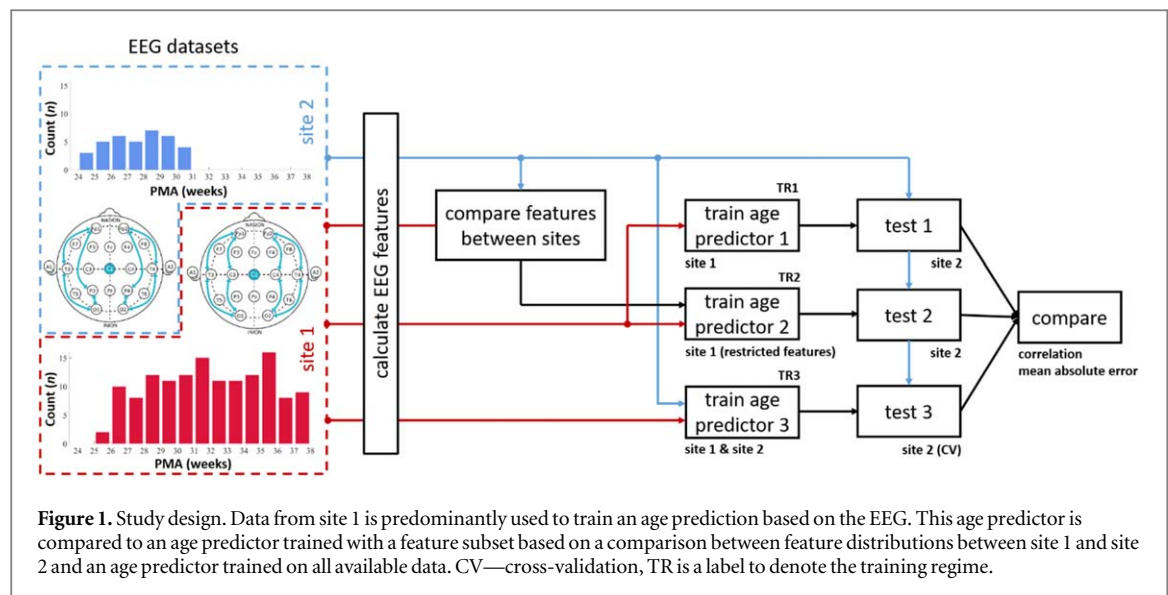
E-mail: nathan.stevenson@qimrberghofer.edu.au**Keywords:** electrophysiology, electroencephalogram, brain age prediction, preterm infants**Abstract**

Objective. To overcome the effects of site differences in EEG-based brain age prediction in preterm infants. **Approach.** We used a 'bag of features' with a combination function estimated using support vector regression (SVR) and feature selection (filter then wrapper) to predict post-menstrual age (PMA). The SVR was trained on a dataset containing 138 EEG recordings from 37 preterm infants (site 1). A separate set of 36 EEG recordings from 36 preterm infants was used to validate the age predictor (site 2). The feature distributions were compared between sites and a restricted feature set was constructed using only features that were not significantly different between sites. The mean absolute error between predicted age and PMA was used to define the accuracy of prediction and successful validation was defined as no significant differences in error between site 1 (cross-validation) and site 2. **Main results.** The age predictor based on all features and trained on site 1 was not validated on site 2 ($p < 0.001$; MAE site 1 = 1.0 weeks, $n = 59$ versus MAE site 2 = 2.1 weeks, $n = 36$). The MAE was improved by training on a restricted features set (MAE site 1 = 1.0 weeks, $n = 59$ versus MAE site 2 = 1.1 weeks, $n = 36$), resulting in a validated age predictor when applied to site 2 ($p = 0.68$). The features selected from the restricted feature set when training on site 1 closely aligned with features selected when trained on a combination of data from site 1 and site 2. **Significance.** The ability of EEG classifiers, such as brain age prediction, to maintain accuracy on data collected at other sites may be challenged by unexpected, site-dependent differences in EEG signals. Permitting a small amount of data leakage between sites improves generalization, leading towards universal methods of EEG interpretation in preterm infants.

Introduction

Differences between physiological derived estimates of age and actual age are intuitive biomarkers of human health (Jylhävä *et al* 2017). These measures have applications in paediatrics where maturation is sufficiently rapid to detect clinically meaningful changes that can be used for diagnosis and prognosis. The rate of maturation in brain function is particularly clear in preterm infants where recent studies on estimates of brain age have showed associations between EEG derived age and neurodevelopment (Stevenson *et al* 2020, Ansari *et al* 2023).

Clinical application of these methods requires generalizability, which can be defined as consistent prediction output across larger unseen populations. Several methods have been proposed and shown excellent consistency when



applied to unseen data recorded at the same center/site (O'Toole *et al* 2016, Stevenson *et al* 2020, Ansari *et al* 2023). However, consistent performance in age estimates based on EEG data from independent sites/centers has not been comprehensively assessed; findings from MRI-based estimates of age suggest considerable reduction in accuracy (Jirsaraie *et al* 2023). Inter-site complications or confounders that may affect the accuracy of EEG-based age prediction algorithms include different types of recording electrodes (e.g. subdermal electrodes versus Ag/AgCl scalp electrodes), methods of electrode attachment (gel/tape, caps), electrode positions (full 10–20, 10–20 modified for neonates, limited channel monitoring), acquisition amplifiers/filter settings, etc. A universal algorithm needs to be based on methods that are invariant to these potential differences, resulting in high generalizability and successful clinical translation.

An apparent solution to improve generalizability is to compile large multicenter datasets to be used for algorithm training. This is often difficult to achieve, particularly retrospectively, as privacy regulations and legal restrictions differ from country to country. There are also other challenges related to different perspectives on using cohort data and limited funding cycles within larger research programs. Methods such as federated learning have been developed to overcome issues with raw data sharing (Rieke *et al* 2020, Borovac *et al* 2022). These methods typically involve sharing certain, unidentifiable aspects of the data only, to improve algorithm consistency or generalizability across the wider population. Although they are generally designed for methods of learning with deep neural networks, several tenets hold for 'bag of features' based supervised learning. Indeed, sharing only specific aspects of data is well suited to 'bag of features' based supervised learning as it is sufficient to only share the extracted features rather than the raw data.

In this note, we investigated a key challenge in EEG based age prediction in preterm infants; site differences between EEG features. We show that generalization of age prediction across sites was improved by using a subset of features that were similarly distributed with respect to age between sites. This small leakage of data between training and validation sites has the potential to accelerate algorithm development through efficient use of multi-site data collection.

Methods

The general study design is shown in figure 1. We compare the accuracy of three age predictors using EEG recordings from two independent sites. The three age predictors use (1) training data from only site 1 (TR1); (2) training data from only site 1 with pruning of the original feature set to only include features that are comparably distributed between sites (TR2) and (3) training data from site 1 and site 2 (TR3). The consistency of algorithm performance is defined by comparing measures of accuracy such as correlation and mean absolute error between post-menstrual age (PMA) and age prediction between training (evaluated using cross-validation) and testing/validation (site 2) datasets.

Dataset 1 (site 1)

The dataset collected from site 1 consisted of 67 preterm infants admitted to the NICU at the Vienna General University Hospital, Austria (sex: 34 males, 31 females, 2 data missing). Infants were included if they were born before 29 weeks gestational age, medically stable at the time of EEG recordings, and parental consent was received. EEG was acquired as soon as possible after birth and then at fortnightly intervals, whenever possible, using a Brain Quick / ICU

EEG (MicroMed, Treviso, Italy) with a sampling frequency of 256 Hz. EEG recordings were generally acquired between 9 am and 5 pm following cares and between feeds. Electrodes were located according to the 10–20 international system (modified for neonates): Fp1, Fp2, C3, C4, T3, T4, O1, O2, with a reference at Cz. A bipolar montage (double banana) was then used in analysis: Fp1–C3, C3–O1, Fp1–T3, T3–O1, Fp2–C4, C4–O2, Fp2–T4, T4–O2. Recordings contained at least one complete cycle of sleep and were on average 2.6 h long. All infants had neuropsychological testing at 2 years of age with Bayley's Scales of Neurodevelopment (II and III; testing transitioned during the study period), we selected infants with the lowest score across all domains greater than 70 as these have been shown to not confound age prediction based on EEG analysis (Stevenson *et al* 2020). This resulted in 138 recordings from 37 infants (sex: 13 males, 24 females) and 59 recordings from 37 infants in the age range of 24–31 weeks (comparable to site 2) with a post-natal age (PNA) of 23 d (due to multiple recordings within this time period). The imbalance in sex within a subgroup that excludes poor outcomes reflects the male disadvantage seen in neonatal outcomes (Naeye *et al* 1971). Of these infants, 19 had mildly abnormal neurodevelopment outcome (minimum score between 70 and 85) and 9 were SGA. More details of this dataset can be found in Stevenson *et al* (2020). The original data collection was approved by the local ethics committee (Medical University Vienna, Austria) and the protocol was registered (Ethics no. 1814/2018).

Dataset 2 (site 2)

The dataset collected from site 2 consists of 41 preterm infants (sex: 22 males, 19 females) admitted to the NICU at Ullevaal, Oslo University Hospital, Norway. Infants with gestational age 24^{+0} to 30^{+6} weeks and no congenital abnormalities were included. EEG monitoring started within 12 h of life, and the infants were monitored until 72 h of life. We extracted a 24 h period of EEG recording commensurate with day 2 of life as it is during this period that minimal disruption from clinical care was apparent and EEG analysis has the highest correlation with health outcomes (Nordvik *et al* 2022). The EEG recording was performed using NicoletOne™ version 5.2 EEG system (Natus Medical Inc., Middleton, USA) with a sampling frequency of 256 Hz and were on average 23 h long. We used 8 electrodes and 2 reference electrodes that were positioned according to the modified version of the 10–20 international system: Fp1, Fp2, P3, P4, T3, T4, O1 and O2. For analysis, a bipolar montage similar to the Vienna cohort (using parietal electrodes instead of central electrodes) was used: Fp1–P3, P3–O1, Fp1–T3, T3–O1, Fp2–P4, P4–O2, Fp2–T4 and T4–O2. The children were assessed at 2 years corrected age using Bayley Scales of Neurodevelopment II. Similar to site 1, infants with a neurodevelopmental score less than 70 in any domain were excluded, resulting in 36 recordings from 36 infants (sex: 20 males, 16 females) recorded at an average PNA of 2 d. Of these infants, 18 had mildly abnormal neurodevelopmental outcome and 9 were SGA. More details of this dataset can be found in (Schumacher *et al* 2011). The original data collection was approved by the Regional Committee for Medical Ethics (REK 2011/1214) and the Norwegian Data Protection Supervisor.

EEG features

Summary measures of the EEG (features) were designed to capture important signal characteristics in the time, frequency and information domains. A total of 43 features were calculated on the EEG (see table A1 for a full list of the features). This is smaller than the original feature set used in Stevenson *et al* (2020), as cross-channel features were removed to ensure montage agnostic performance and several burst metrics were removed or rewritten to ensure accessibility in the public domain (see <https://github.com/nstevensonUH/Neonatal-EEG-Analysis> for a public repository of Matlab code to calculate the features). EEG features were calculated on a 1 h epoch of EEG, resampled to 64 Hz. EEG features were calculated on each recording channel and the median feature value across channels was used to summarize the EEG epoch. Each recording was, therefore, represented by a feature vector of size $N \times M$, where N is the number of EEG recordings in the dataset and $M = 43$ is the number of features. If more than one 1 h epoch is extracted from a recording the median feature value across all EEG epochs within a single recording is used. The electrode locations are slightly different between site 1 and site 2 (Central versus Parietal). We assume these differences in electrode positions will not significantly affect the global average due to the small head size minimizing the distance between central and parietal locations and our use of averaging across electrodes to form a global feature estimate.

Feature comparison

While EEG recordings from site 1 had a wider range of PMA (23–38 weeks PMA), feature comparisons were limited to a PMA of less than 31 weeks to ensure similar age ranges between site 1 and 2. To perform feature comparisons, feature values from both sites were combined and then corrected for any relationship with age using polynomial fitting (polynomial order from 1 to 3 was selected using the Akaike Information Criteria (AIC)). Site differences were then compared using statistical hypothesis testing (unpaired t-test) where the null hypothesis was that there was no difference in EEG feature between sites. The correlation between individual features and age per site was also computed. Features that were not significantly different were included in a

reduced, site agnostic, feature set. In this case, the level of significance was 0.05 and no correction for multiple comparisons were used resulting in more conservative feature selection (i.e. a smaller feature set).

Age prediction

Age prediction was based on the combination of several features extracted from the 1 h epochs of EEG. The combination function was estimated via a training procedure within a support vector regression (SVR) (Stevenson *et al* 2020). Feature selection was applied to all training regimes, used to reduce the dimensionality of the input feature vector and performed using a combined filter-wrapper approach. Features were initially selected based on their correlation with age; only features with a significant correlation (corrected for multiple comparisons; Benjamini–Hochberg procedure) with age were included (filter stage). The residual feature set was applied to a backwards feature selection with a stopping criteria based on the AIC evaluated on an internal cross-validation within a training set iteration (wrapper stage). The implementation of the filter stage was based on the assumption that potential improvements in prediction due to features uncorrelated with age were minor and most susceptible to inter-site differences, i.e. we select only universally relevant features. The wrapper stage eliminates redundant features. Training used the entire age range of the available data. Cross-validation was used to determine within dataset performance. The SVR was trained using the Matlab function `fitsvm.m` with a medium Gaussian kernel (Kernel Scale = 9.8, box constraint = $\text{IQR}/1.349$, $\varepsilon = \text{IQR}/13.49$ and IQR is the interquartile range the input PMA).

Experiments

Three age predictions were then compared to examine the effects of feature selection strategies to overcome site differences: (1) an age prediction trained entirely on site 1 (TR1; feature selection within a four-fold cross-validation and tested on site 2); (2) an age prediction trained entirely on site 1, with a reduced feature set consisting of features that behave similarly between site 1 and site 2 (TR2; feature selection within a four-fold cross-validation and tested on site 2); and (3) an age prediction trained on a combined site 1 and site 2 dataset (TR3; feature selection within a four-fold cross-validation). All training regimes use the hybrid filter-wrapper feature selection approach. The effectiveness of these methods was evaluated by comparing age prediction accuracy on a validation set (site 2).

The accuracy of age prediction was defined using mean absolute error (the absolute value of predicted age minus PMA averaged across all EEG recordings), bias (the predicted age minus PMA averaged across all EEG recordings) and the linear correlation coefficient (Pearson's). We defined validation by comparing the absolute error of the age prediction estimated on site 1 (via cross-validation) and site 2. We used an unpaired t-test (one-sided, level of significance is 0.05) with a linear correction for age (the error/residual is proportional to age). If no differences in residuals between site 1 and site 2 were found then the age prediction algorithm was assumed to be validated. We also observed the features selected across the various feature selection strategies to ensure that the proposed strategy (TR2) approaches the ideal case (TR3). Ideally, the features selected using the reduced feature set should converge to the features selected using the combined site 1 and site 2 training data. All comparisons were performed on data with a similar age range (site 1: $n = 37$, site 2: $n = 36$).

Results

Features had varying degrees of similarity across sites and correlation with PMA (see figure 2). In summary, 29/43 (67%) of EEG features were not significantly different between sites after correcting for a polynomial relationship between feature and PMA and multiple comparisons. Of these 29 EEG features, 15 (52%) had a correlation that was significantly different from zero. These measures include EEG amplitude, burst sharpness (over several time scales defined by burst duration), slopes/intercept of burst duration versus burst area/sharpness, suppression curve, burst duration (mean and standard deviation), relative alpha power and sample entropy (these features are highlighted in table A1).

Eliminating features that were dissimilar between sites and retraining an age predictor based on data only from site 1, improved cross-validation accuracy within site 1 and considerably improved the MAE of the age predictor when applied to independent data in site 2 (see table 1). Training regimes 2 and 3 (TR2 and TR3) resulted in a validated age prediction algorithm (no significant difference between site 1 and site 2 residuals). This improvement in validation accuracy was achieved with less features in the age prediction algorithm from TR1 and was primarily achieved by improving the bias between predicted age and PMA. As a result, the age prediction based on site 1 with the reduced feature set (TR2) had an error that was not significantly different from a subgroup with similar age group from site 1 when adjusted for age effects on error (site 1: mean adjusted error = -0.03 weeks, standard deviation = 1.11 weeks; $n = 59$ and site 2: mean adjusted error = 0.05 weeks, standard deviation = 0.73 weeks; $n = 36$, $p = 0.68$). Further analysis showed that the initial pruning of the feature set resulted in selections that were better aligned with those obtained after feature selection on a combined site 1/site 2 dataset (TR3)—see figure 3. The prediction of age from TR2 for site 2 also had an improved correlation between prediction and age, similar MAE but

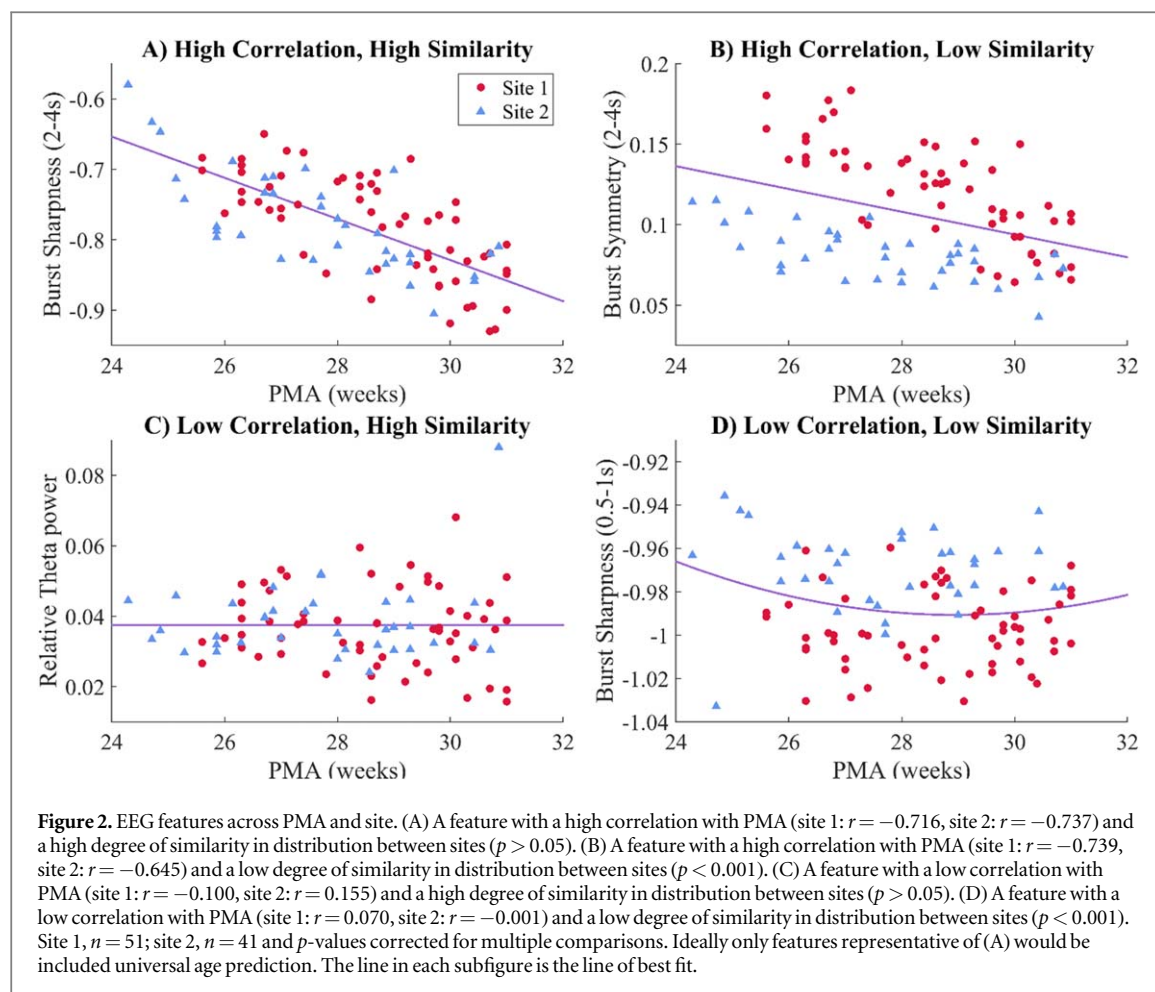


Table 1. Cross-validation and validation accuracy of age prediction using different training data. No FS—full feature set with no feature selection. FFS—full feature set with feature selection, RFS—restricted feature set with feature selection, CV—cross-validation. Feature selected is the median number of features selected per CV fold. TR denotes the training regime number with respect to figures 1 and 3.

Training data	Features selected	Testing: CV site 1		Testing: site 2		
		r^2	MAE (weeks)	r^2	MAE (weeks)	Bias (weeks)
Site 1 (no FS)	43	0.877	0.9	0.613	2.2	-2.1
Site 1 (FFS:TR1)	8	0.849	1.0	0.736	2.1	-2.1
Site 1 (RFS:TR2)	6	0.861	1.0	0.709	1.1 ^a	-0.9
Site 1 + site 2 (FFS:TR3)	12	0.849	1.1	0.737*	0.7 ^{*,a}	-0.3*

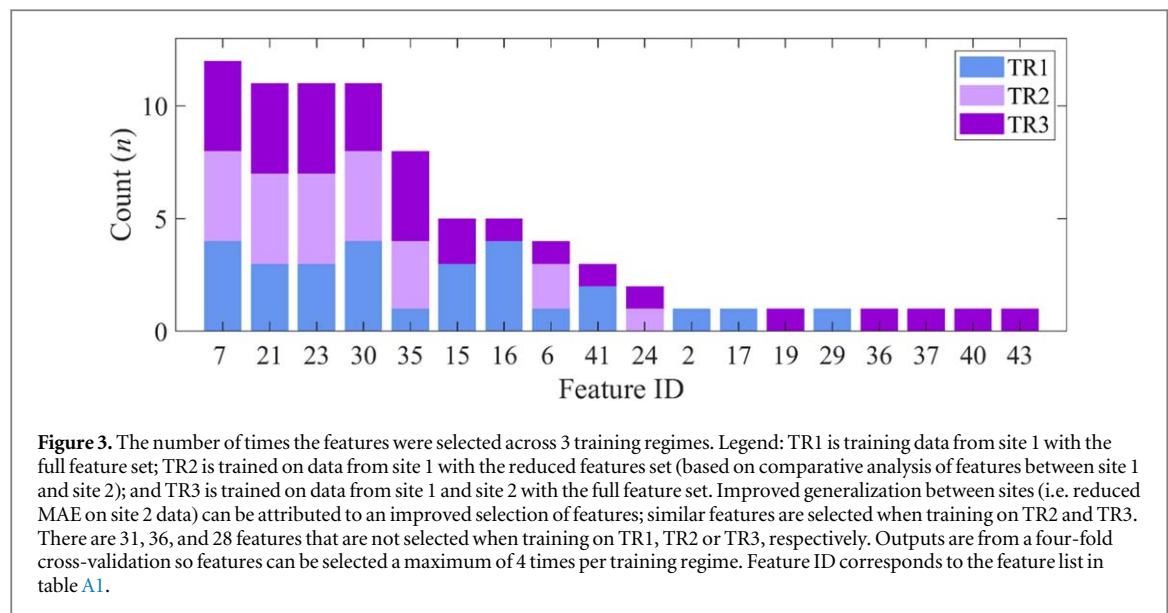
*Cross-validation.

^a No significant difference between site 1 and site 2.

increased bias, compared to a prediction of age based on training on only site 2 ($r^2 = 0.498$, MAE = 1.04 weeks, bias = -0.1 weeks; 10-fold cross-validation using only site 2 data, $n = 36$). There was no effect on biological sex on prediction error for site 2 data across all training regimes ($p = 0.56$, $p = 0.47$, $p = 0.98$, for TR1, TR2, and TR3, respectively: Mann-Whitney U test, $n = 36$).

Discussion

Our results show that the EEG recording site has significant impacts on EEG feature values despite similar clinical characteristics between cohorts. Approximately one third of our features had significantly different age-dependent distributions between the two study sites. This disparity leads to poor validation performance when training on one site and testing on the other. Validation performance can be improved by eliminating features



with dissimilar distributions between sites, *a priori*. The general implication of this study is that individual features within larger sets must be scrutinized not only for their representation of key physiological functions but also inter-site consistency if clinical translation is a priority.

We show clear differences in the maturation of EEG features between sites. While the differences in features of similar characteristics (i.e. burst measures, amplitude measures) was somewhat inconsistent (see table A1), the majority of differences appears to be in the frequency domain with deviations in the higher frequencies of recorded activity (e.g. relative beta power and spectral slope). We speculate two main contributions to site differences: (1) the PNA of recordings and (2) differences in the frequency response of recordings from the specific combination of electrode type/fixation/EEG amplifier. We have shown that early post-natal adaption effects the maturation of EEG features (O'Toole *et al* 2019) and all site 2 data is in within a period of early post-natal adaption (mean PNA = 2 d) while site 1 data is less likely to fall in this period (mean PNA = 23 d). We suspect the effect of post-natal adaption is greater than other differences such as the length of recordings which would only affect the variance of the EEG feature not the systematic bias seen in some features (e.g. figure 2(B)). Any differences in EEG technology/implementation will also affect the frequency content and, therefore, the signal to noise ratio of the EEG recording (O'Sullivan *et al* 2019). Nevertheless, there remain features that are invariant to these effects and, in combination, provide an accurate estimate of age across sites, negating the need for additional normalization stages to account for site differences; a unnecessarily complicated solution.

In a standard machine learning paradigm, it is well known that sharing or leaking data between training and testing/validation sets results in an overestimate of performance when applied to data collected at independent sites (Kaufman *et al* 2012, Kapoor and Narayanan 2022). The hard criterion of not sharing data (zero leakage), although theoretically ideal, is problematic as validation performance can only be determined when tested, leading to a cycle of intra-site cross-validation, inter-site validation, failure, and additional data collection. The obvious solution is to initially collect large multi-site datasets, but this is often not viable as data sharing is hampered by legal constrictions, while large scale data collection is inherently time-consuming and expensive; therefore, not easily supported by typical funding instruments. We addressed this issue by breaking the theoretical hard criterion; comparing features across sites and eliminating features that were not similarly distributed over the age range of interest. This meant that improvements to the age prediction model from access to a larger training set were maintained across sites, i.e. we showed increased accuracy from a model trained on site 1 ($n = 138$) and applied to site 2 ($n = 36$) compared to model trained and tested (cross-validation) on site 2 (r^2 increased from 0.498 to 0.709). The use of inter-site feature screening also improved validation accuracy with no difference in residuals between site 1 and site 2 and approached a level of accuracy of an algorithm trained and cross-validated on a combined site dataset (TR3). The accuracy of the algorithm was similar to other age prediction methods with MAE values of 0.7 weeks reported (Stevenson *et al* 2020, Ansari *et al* 2023), although it is often difficult to compare these values due to differences in cohort demographics, age ranges and EEG acquisition environments. It is also worth noting the differences in our approach to methods such as transfer learning (Rieke *et al* 2020). In our approach we used a low level of data leakage to train a single age prediction algorithm that has consistent performance across multiple sites, while traditional transfer learning provides a method of re-training the last layers of a network, pre-trained on data from site 1, using a small amount of data from site 2. Transfer learning is also only applicable to deep neural networks and as brain age estimation in

preterm infants is in a nascent phase, feature based methods are better positioned to help understand effects of potentially important clinical factors such as outcome, medication, and biological sex, on age estimates as feature based methods are more ‘explainable’ than deep neural networks.

There were limitations to this study. The EEG is sensitive to a range of factors including injury, PNA, condition and medication (Bell *et al* 1993, Watanabe *et al* 1999, Hayashi-Kurahashi *et al* 2012, O’Toole *et al* 2019). As this work compared EEG recordings between sites, in the context of age prediction, rather than developing the clinical utility of age prediction we erred on the side of more data (intra-cohort heterogeneity) rather than clinical precision (intra-cohort homogeneity). While we excluded infants with clearly abnormal neurodevelopmental outcome according to neuropsychological testing, we did not eliminate infants with only mildly abnormal outcome and/or SGA. We included mildly abnormal infants as they have been shown to not affect age prediction and we included SGA infants as a proportion of these infants can be considered small but normal (we did not discriminate between SGA and intra-uterine growth restriction). However, when translating age predictors to the clinic for use as a biomarker of neurodevelopment, clinical precision of training cohorts becomes paramount. Nevertheless, it is difficult to see how relatively small levels of inter-cohort heterogeneity can explain the magnitude of site differences seen in figures 2(B) and (D). There are notable biological sex differences in the EEG (e.g. frontal EEG) during childhood development with more limited evidence of effects in early infancy (Thordstein *et al* 2006, Gartstein *et al* 2020). We did not build biological sex specific age predictors due to limitations in the sample size and a priority of maximizing available training data. While separate predictors for male and female may be useful and align with other measures of developmental growth, they may not be necessary for global assessment of functional age at such an early time point in life, particularly as machine learning methods are capable of correcting for any difference in EEG attributable to biological sex in pursuit of maximum age prediction accuracy. This may explain why biological sex had no effect on prediction error in our study, not because differences do not exist but rather were corrected for during training.

Bag of feature based pattern recognition systems have been surpassed in classification accuracy by advances in deep learning. There are, however, specific applications where they retain advantages. We show that *a priori* comparison of EEG feature distribution between sites (a form of data leakage) improves the validation performance of age prediction in preterm infants. This has the potential to accelerate the cycle of development, validate, development commonly seen when developing medical AI tools. The effects of this data leakage are assumed to be negligible but must be further tested to ensure validity.

Acknowledgments

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Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. Features derived from the EEG that support the findings of this study are available upon reasonable request from the authors.

Ethical statement

Our research was conducted in accordance with the principles embodied in the Declaration of Helsinki and in accordance with local statutory requirements. The data collection from site 1 was approved by the local ethics committee (Medical University Vienna, Austria) and the protocol was registered (Ethics no. 1814/2018). The data collection from site 2 was approved by the Regional Committee for Medical Ethics (REK 2011/1214) and the Norwegian Data Protection Supervisor. For both datasets, informed, written, parental consent was received before data collection commenced.

Appendix

Table A1. Comparing features between sites and with PMA. Only feature values between the ages of 23 weeks to 31 weeks were analyzed. r is the linear correlation coefficient between feature and age. The p -value is the output of a t-test of feature values between sites corrected for age using a low order polynomial (1st, 2nd or 3rd order, depending on best fit as assessed by AIC), note the p -value is uncorrected for multiple comparisons.

	Feature name	Site 1 r	Site 2 r	p -value
1	Median amplitude envelope	0.220	0.482	0.001
2	5th percentile amplitude envelope*	0.260	0.585	0.004
3	25th percentile amplitude envelope*	0.266	0.545	0.002
4	75th percentile amplitude envelope*	0.128	0.328	0.002
5	95th percentile amplitude envelope	0.029	−0.109	0.000
6	5th percentile inter-burst interval*	0.043	0.533	0.297
7	50th percentile inter-burst interval*	−0.165	−0.023	0.008
8	95th percentile inter-burst interval*	0.146	−0.021	0.066
9	5th percentile burst duration*	0.259	0.687	0.002
10	50th percentile burst duration*	0.211	0.408	0.811
11	95th percentile burst duration*	0.069	0.285	0.480
12	Burst symmetry (15.625–125 ms)*	−0.137	−0.151	0.425
13	Burst symmetry (125–250 ms)*	−0.169	−0.543	0.012
14	Burst symmetry (250–500 ms)	−0.192	−0.655	0.000
15	Burst symmetry (0.5–1 s)	−0.259	−0.594	0.000
16	Burst symmetry (1–2 s)	−0.557	−0.646	0.000
17	Burst symmetry (2–4 s)	−0.760	−0.647	0.000
18	Burst symmetry (15.625 ms–128 s)	−0.600	−0.622	0.000
19	Burst sharpness (15.625–125 ms)	−0.058	−0.222	0.000
20	Burst sharpness (125–250 ms)*	0.349	0.611	0.217
21	Burst sharpness (250–500 ms)*	−0.608	−0.553	0.656
22	Burst sharpness (0.5–1 s)	0.046	−0.001	0.000
23	Burst sharpness (1–2 s)*	0.008	−0.107	0.824
24	Burst sharpness (2–4 s)*	−0.713	−0.737	0.059
25	Burst sharpness (15.625 ms–128 s)*	−0.559	−0.638	0.326
26	Slope burst duration versus burst area*	−0.631	−0.688	0.087
27	Intercept burst duration versus burst area*	0.258	0.512	0.003
28	Slope burst duration versus skewness	−0.718	−0.594	0.000
29	Intercept burst duration versus skewness	−0.296	−0.613	0.000
30	Slope burst duration versus kurtosis*	−0.676	−0.721	0.103
31	Intercept burst duration versus kurtosis*	0.224	0.266	0.514
32	Suppression curve*	−0.472	−0.574	0.861
33	Mean burst duration*	−0.595	−0.482	0.648
34	Standard deviation burst durations*	−0.561	−0.649	0.227
35	Sample entropy*	0.599	0.661	0.026
36	Log PSD energy (wideband)	−0.039	−0.046	0.001
37	Relative delta 1 power*	−0.022	−0.293	0.545
38	Relative delta 2 power*	−0.109	−0.155	0.119
39	Relative theta power*	−0.088	0.155	0.438
40	Relative alpha power*	0.269	0.493	0.012
41	Relative beta power	0.372	0.655	0.000
42	Spectral slope	0.249	0.589	0.001
43	Spectral edge frequency (90th percentile)*	0.083	0.311	0.290

*Features included in the reduced feature set (also italicized).

References

- Ansari A *et al* 2023 Brain Age as an Estimator of Neurodevelopmental Outcome: a Deep Learning Approach for Neonatal Cot-side Monitoring. *bioRxiv*.2023-01
- Bell A H, Greisen G and Pryds O 1993 Comparison of the effects of phenobarbitone and morphine administration on EEG activity in preterm babies *Acta Paediatrica* **82** 35–9
- Borovac A, Gudmundsson S, Thorvardsson G, Moghadam S M, Nevalainen P, Stevenson N, Vanhatalo S and Runarsson T P 2022 Ensemble learning using individual neonatal data for seizure detection *IEEE J. Trans. Eng. Health Med.* **10** 1–1
- Gartstein M A, Hancock G R, Potapova N V, Calkins S D and Bell M A 2020 Modeling development of frontal electroencephalogram (EEG) asymmetry: sex differences and links with temperament *Developmental Sci.* **23** e12891

- Hayashi-Kurahashi N, Kidokoro H, Kubota T, Maruyama K, Kato Y, Kato T, Natsume J, Hayakawa F, Watanabe K and Okumura A 2012 EEG for predicting early neurodevelopment in preterm infants: an observational cohort study *Pediatrics* **130** e891–7
- Jirsaraie R J, Kaufmann T, Bashyam V, Erus G, Luby J L, Westlye L T, Davatzikos C, Barch D M and Sotiras A 2023 Benchmarking the generalizability of brain age models: challenges posed by scanner variance and prediction bias *Human Brain Mapp.* **44** 1118–28
- Jylhävä J, Pedersen N L and Hägg S 2017 Biological age predictors *EBioMedicine* **21** 29–36
- Kapoor S and Narayanan A 2022 Leakage and the reproducibility crisis in ML-based science arXiv:2207.07048
- Naeye R L, Burt L S, Wright D L, Blanc W A and Tatter D 1971 Neonatal mortality, the male disadvantage *Pediatrics* **48** 902–6
- Kaufman S, Rosset S and Perlich C 2012 Leakage in data mining: formulation, detection, and avoidance *Proc. 17th ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining—KDD’11* p 556
- Nordvik T, Schumacher E M, Larsson P G, Pripp A H, Løhaugen G C and Stiris T 2022 Early spectral EEG in preterm infants correlated with neurocognitive outcomes in late childhood *Pediatr. Res.* **92** 1132–9
- O’Sullivan M, Temko A, Bocchino A, O’Mahony C, Boylan G and Popovici E 2019 Analysis of a low-cost EEG monitoring system and dry electrodes toward clinical use in the neonatal ICU *Sensors* **19** 2637
- O’Toole J M, Boylan G B, Vanhatalo S and Stevenson N J 2016 Estimating functional brain maturity in very and extremely preterm neonates using automated analysis of the electroencephalogram *Clin. Neurophysiol.* **127** 2910–8
- O’Toole J M, Pavlidis E, Korotchkova I, Boylan G B and Stevenson N J 2019 Temporal evolution of quantitative EEG within 3 days of birth in early preterm infants *Sci. Rep.* **9** 4859
- Rieke N *et al* 2020 The future of digital health with federated learning *NPJ Digit. Med.* **3** 119
- Schumacher E M, Westvik A S, Larsson P G, Lindemann R, Westvik J and Stiris T A 2011 Feasibility of long-term continuous EEG monitoring during the first days of life in preterm infants: an automated quantification of the EEG activity *Pediatric Res.* **69** 413–7
- Stevenson N J, Oberdorfer L, Tataranno M L, Breakspear M, Colditz P B, de Vries L S, Benders M J, Klebermass-Schrehof K, Vanhatalo S and Roberts J A 2020 Automated cot-side tracking of functional brain age in preterm infants *Annals of Clin. Transl. Neurol.* **7** 891–902
- Thordstein M, Löfgren N, Flisberg A, Lindecrantz K and Kjellmer I 2006 Sex differences in electrocortical activity in human neonates *Neuroreport* **17** 1165–8
- Watanabe K, Hayakawa F and Okumura A 1999 Neonatal EEG: a powerful tool in the assessment of brain damage in preterm infants *Brain Dev.* **21** 361–72